

Entropy-Based Curriculum Training for Land-Cover Semantic Segmentation

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ABSTRACT

Rapid urbanization and environmental changes have made it increasingly important to monitor how land is utilized over time. Traditional land classification methods are often time-consuming, costly and rely heavily on manual interpretation. In this study, we employ a state-of-the-art semantic segmentation model DeepLabV3+ on high-resolution satellite imagery and introduce an Entropy-Based Curriculum Training (EBCT) strategy that gradually exposes the network to samples from low-entropy to high-entropy regions. We investigate whether EBCT improves boundary delineation and class balance compared to standard training protocols and aim to provide an efficient pipeline for land-cover mapping. Experiment results indicate that the proposed EBCT strategy yields slightly better segmentation performance compared to the baseline, while reducing computational costs during training. Across the DeepGlobe and LoveDA benchmarks, EBCT improves mIoU by $\approx 1\%$ over standard training while reducing training-time image computations by up to 91.9% ($\approx 8\%$ additional overhead for entropy estimation).

1. INTRODUCTION

Land use refers to the modifications made by humans to the living environment, such as fields, agriculture and settlements (Manakane et al., 2023; Talukdar et al., 2020). It encompasses both the physical land cover of the Earth’s surface and the way that cover is utilized within a given area (Rakuasa & Pakniany, 2022; Rakuasa et al., 2023). While information on how land is used is distinct from land-cover information, the latter can be recognized directly from remote-sensing satellite images and serves as a primary proxy for understanding land utilization (Achmadi et al., 2023; Rakuasa, 2022). Reliable classification plays an important role in managing resource decline and mitigating the impact of population growth: as urbanization accelerates, effective mapping of human activities on the landscape helps address negative consequences for energy and natural resources, enabling sustainable planning, resource management, and environmental conservation (Avtar et al., 2019).

Recent accessibility to high-resolution remote-sensing images and the ability to gather multi-temporal and multi-source RS (remote-sensing) images from diverse geographic regions (Sun et al., 2019) present new opportunities for multiple-time-scale classification tasks. Nevertheless, the complex features visible in HRRS images, such as geometrical and object

structures, introduce new challenges for classification (Shiran et al., 2019). Variability in photographic distortions, scale variations and illumination changes in RS images makes it difficult to apply existing models to diverse HRRS images effectively. The core problem addressed in this research is the “semantic gap” introduced by high-resolution imagery. Furthermore, spectral and spectral-spatial (SS) features have traditionally been employed in the literature to interpret RS images for land-cover classification (Alem & Kumar, 2022). However, these features struggle to capture the contextual information in high-resolution images due to increased spatial resolution (Alrayes et al., 2022). To address these challenges, deep CNNs have gained attention for their ability to comprehend HRRS images (Tong et al., 2020) and represent semantic and high-level image properties (Zhao et al., 2020) such as scene classification, object detection, image retrieval as well as LC classification.

To better exploit this heterogeneous data, we explore an Entropy-Based Curriculum Training (EBCT) strategy. Instead of sampling all training patches uniformly, EBCT orders samples from “easy” to “hard” using the entropy of the model’s predicted class distribution as a proxy for uncertainty. The network is first trained on low-entropy tiles and is gradually exposed to high-entropy regions such as mixed urban–vegetation boundaries and visually ambiguous areas, with the hypothesis that this curriculum can stabilize optimization and sharpen segmentation boundaries.

In summary, we focus on the problem of accurately classifying key land-cover categories (e.g., urban, agricultural, and forest areas) from high-resolution satellite imagery. Our objectives are (i) to build a strong deep-learning baseline using recent semantic segmentation models (Cao et al., 2021; Chen et al., 2018), and (ii) to investigate whether the proposed EBCT scheme can further improve their performance by exploiting the inherent curriculum structure of the data. This work aims to provide a small but concrete methodological contribution to the land-cover classification literature by showing how a simple entropy-based curriculum can be integrated into existing state-of-the-art models.

2. BACKGROUND AND RELATED WORK

Land-use and land-cover (LULC) information is a key input for urban planning, agricultural monitoring, and environmental management, especially under rapid urban expansion and climate change. High-resolution satellite imagery enables detailed mapping of cropland, forest, and built-up areas and supports long-term analyses of land-use change at local and regional scales (Rakuasa & Achmadi, 2024; Tamás et al., 2025). Recent studies highlight how accurate LULC maps directly feed into applications such as water-resource management, food-production forecasting, and the assessment of human pressure on ecosystems (Tamás et al., 2025).

The literature reveals a clear trajectory in remote sensing methodologies. Early approaches utilized feature engineering (like GLCM textures) coupled with classifiers like Random Forest

(RF) or Support Vector Machines (SVM). While RF remains computationally efficient and resistant to overfitting, it lacks the capacity to model spatial context. The field subsequently shifted toward Convolutional Neural Networks (CNNs) like U-Net (Ronneberger et al., 2015) and SegNet (Badrinarayanan et al., 2017), which improved accuracy by learning hierarchical features. Most recently, the focus has moved to DeepLabV3+ and Transformer-based models like Swin-UNet (Cao et al., 2021) which have demonstrated enhanced performance in capturing long-range dependencies and contextual information, making them highly effective for high-resolution land-cover classification tasks.

Early work on high-resolution LULC mapping relied on hand-crafted spectral and textural features combined with classical classifiers systematically evaluated different feature sets and classifiers for urban land-use / land-cover mapping from very-high-resolution imagery, showing that careful feature selection together with Random Forest or SVM can reach F1-scores around 0.90 on complex urban scenes. Similar ideas appear in regional water-resource studies, where multi-sensor Landsat-8 and Sentinel-2 data are fused and then classified with conventional machine-learning pipelines to derive land-cover maps that drive hydrological and resource-management models (Tamás et al., 2025). While these approaches are relatively efficient and interpretable, they treat each pixel or object largely independently and struggle to encode the rich spatial context visible in sub-meter imagery.

Motivated by these limitations, recent work has shifted toward deep learning on high-resolution remote-sensing (HRRS) imagery. Hao et al. (2024) investigated several deep models for LULC classification from HRRS images and demonstrated clear gains over traditional feature-engineering pipelines. Fayaz et al. (2024) fine-tuned Inception-v3 and DenseNet-121 on the UC Merced Land Use dataset and reported around 92% overall accuracy, illustrating the effectiveness of transfer learning for scene-level land-use classification with limited labeled samples. These works show that convolutional networks can learn more discriminative spectral-spatial representations than hand-crafted descriptors, but they primarily operate at the patch or scene level, which limits their ability to precisely delineate class boundaries within heterogeneous tiles.

To obtain pixel-level maps, many recent studies adopt semantic-segmentation architectures. Deressu et al. (2025) proposed AE-DeepLabV3+, an enhanced DeepLabV3+ model with an Xception backbone for land-use and land-cover segmentation, and showed that it outperforms vanilla DeepLabV3+ on a multi-class benchmark including built-up, roads, agriculture, shrubland, forest, grassland, and water. Wang et al. (2024) introduced DeepGDLE, a lightweight variant of DeepLabV3+ that replaces the standard backbone with GhostNet and uses depthwise separable convolutions and channel attention to reduce parameters while improving mIoU on datasets such as WHDLD and FRSID. Together, these studies indicate that both “heavy” and “lightweight” DeepLab-style models can produce accurate, fine-grained LULC maps, and that

architectural choices (backbone, attention, loss) have a significant impact on performance and efficiency.

Other works focus on temporal dynamics and change detection. Rakuasa and Achmadi (2024) conducted a spatial analysis of land-use change between 2014 and 2024 using high-resolution satellite imagery, linking land-cover transitions to urban expansion and policy decisions. Tamás et al. (2025) similarly used multi-temporal land-cover maps as inputs to regional water-resource modelling, showing how classification errors can propagate into downstream decision-making. These studies underline the importance of robust and stable land-cover classifiers, as even modest improvements in accuracy at class boundaries can have amplified effects on long-term environmental assessments.

Across this literature, two patterns are particularly relevant to our work. First, there is a clear evolution from feature-engineering with shallow classifiers (Tamás et al., 2025) to deep classification models (Hao et al., 2024; Fayaz et al., 2024) and finally to segmentation-based architectures tailored for HRRS imagery (Deressu et al., 2025; Wang et al., 2024). Second, almost all deep-learning approaches train their models using a uniform sampling strategy: all training patches are treated as equally important and equally difficult throughout training, even though real land-cover datasets contain a wide spectrum of difficulty, from homogeneous single-class parcels to noisy, mixed urban–vegetation boundaries. To the best of our knowledge, none of the above remote-sensing studies explicitly exploit this difficulty structure via curriculum-style training schedules.

Our study builds on these advances by taking DeepLabV3+ as a strong and widely adopted baseline for high-resolution land-cover segmentation. DeepLabV3+ has demonstrated competitive performance across diverse remote-sensing benchmarks and is also used as the reference architecture in recent variants such as AE-DeepLabV3+ (Deressu et al., 2025) and the lightweight DeepGDLE model (Wang et al., 2024). On top of this standard architecture, we introduce an Entropy-Based Curriculum Training (EBCT) scheme for land-cover classification from high-resolution satellite imagery. By using prediction entropy as a proxy for sample difficulty and gradually moving from low-entropy (easy) tiles to high-entropy (hard) boundary regions, EBCT acts as a simple training-time mechanism that complements existing architectural improvements and directly targets convergence speed, boundary delineation, and class-wise balance.

3. METHODOLOGY

This study leverages the DeepLabV3+ architecture, a state-of-the-art semantic segmentation model, enhanced by Entropy-Based Curriculum Training (EBCT) strategy. By structuring the training process to mimic human cognitive learning, progressing from "easy," low-entropy examples to "hard," high-entropy scenarios, we posit that the model will achieve superior performance in delineating complex boundaries and identifying spectrally ambiguous classes.

This section details the comprehensive experimental design, ranging from the data acquisition and the mathematics of tensor preprocessing to the architectural complexities of neural network architecture.

3.1 DATA AND PREPROCESSING

3.1.1 DEEPGLOBE LAND COVER

The DeepGlobe Land Cover Classification Dataset serves as the primary corpus for training the model on satellite-based, sub-meter imagery. Released as part of the CVPR 2018 DeepGlobe Challenge, it provides a diverse collection of landscapes suited for multi-class semantic segmentation.

A) SPATIAL RESOLUTION

The Ground Sampling Distance (GSD) is 50 cm per pixel. At this resolution, large structural elements (roads, buildings) are clearly resolved, while fine-grained textures (e.g., subtle vegetation patterns or crop-row details) may be aliased, motivating architectures with strong contextual aggregation.

B) IMAGE DIMENSIONS

The training set comprises 803 images, each with a spatial dimension of 2448×2448 pixels. This large tile size provides extensive context but typically requires tiling strategies to fit within GPU memory constraints.

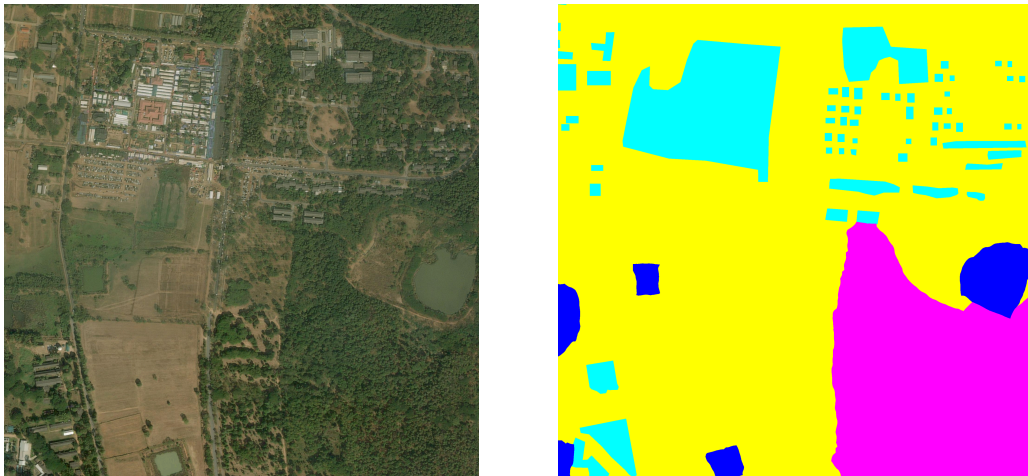


Figure 1: Example from DeepGlobe, an image and its mask.

C) STATISTICAL DISTRIBUTION AND IMBALANCE

A key challenge in DeepGlobe is class imbalance: the dataset is dominated by “Agriculture” in many tiles, while “Urban” and “Water” can be sparse. This motivates the use of imbalance-aware objectives (e.g., Dice-based losses) and curriculum-based sampling to reduce bias toward majority classes. As shown in Table 1, DeepGlobe exhibits severe class imbalance (e.g., Agriculture $\approx 56.8\%$ vs. Water $\approx 3.7\%$), while the Unknown label is negligible ($\approx 0.04\%$) and is treated as an ignored region during optimization and evaluation.

Class	Pixel count (M)	Proportion (%)
Urban	642.4	9.35
Agriculture	3898.0	56.76
Rangeland	701.1	10.21
Forest	944.4	13.75
Water	256.9	3.74
Barren	421.8	6.14
Unknown	3.0	0.04

Table 1: Official pixel-level class distribution in the DeepGlobe Land Cover dataset.

D) PREPROCESSING AND PATCH SAMPLING

The raw dataset consists of high-resolution satellite images paired with RGB label masks. Since the official DeepGlobe validation/test partitions do not provide ground-truth masks, we construct an internal split by sampling **90%** of the labeled images for training and **10%** for validation. Masks are converted from RGB colors to a **single-channel class-index label map** using the known class-color mapping. The *unknown* label is extremely rare ($\approx 0.04\%$ of pixels) and is therefore treated as an **ignored region** during optimization and evaluation.

To make training feasible at 2448×2448 resolution, we sample fixed-size patches of **1024×1024** pixels. Specifically, we draw **two** random patches per image per epoch by sampling crop coordinates on the full tile. In addition to uniform random sampling, we employ a **boundary-guided** patch sampling mechanism to increase the frequency of difficult agriculture-rangeland transitions (which are prone to confusion due to visual similarity, occasional annotation noise, and strong class imbalance). After a short warm-up period, a fixed fraction of patches is selected from regions where rangeland borders agriculture, while the remaining patches are sampled uniformly.

For data augmentation, we apply standard spatial and photometric transforms (random horizontal/vertical flips, 90° rotations, and mild color jitter/brightness-contrast perturbations). Finally, inputs are normalized using ImageNet statistics to match the pretrained encoder and

converted to PyTorch tensors. Each training batch therefore consists of normalized image patches and their corresponding **class-index** masks (with ignored regions).

Note: Same patch extraction and boundary-guided sampling procedure is used for both baseline and EBCT runs; EBCT only changes the curriculum-based subset selection over epochs.

3.1.2 LOVE DA DATASET

The LoveDA (Land-cover Domain Adaptive semantic segmentation) dataset is used as a second benchmark to assess the robustness of our pipeline under domain shifts, leveraging its explicit Urban and Rural scene organization. The dataset is provided as fixed-size image tiles with corresponding semantic masks and is commonly used to evaluate generalization across heterogeneous land-cover patterns.

A) CLASS DISTRIBUTION AND IMBALANCE

Figure 2 summarizes the pixel-level class distribution in LoveDA. The dataset exhibits a noticeable imbalance, with Background and Agricultural occupying the largest pixel share, while minority classes such as Road and Barren are comparatively sparse. This distribution motivates imbalance-aware training choices (e.g., Dice-style objectives) and supports evaluating whether curriculum-based sampling improves performance on underrepresented classes.

B) PREPROCESSING AND CLASS MAPPING

We follow the official dataset structure and construct the training and validation sets by gathering image-mask pairs from the Train/Urban, Train/Rural, Val/Urban, and Val/Rural folders. LoveDA masks are single-channel label maps where 0 denotes no-data; we map this value to an ignored label (255) and shift valid class IDs from 1..7 to 0..6, yielding a contiguous class-index representation. We use the 7-class taxonomy: background, building, road, water, barren, forest, and agriculture.

C) PATCH SAMPLING AND AUGMENTATION

During training, we sample 768×768 patches using RandomCrop, followed by standard spatial augmentations (horizontal/vertical flips and 90° rotations) and mild photometric augmentation (ColorJitter). For validation under a fixed patch protocol, we apply CenterCrop with the same size (768×768) to ensure consistent input dimensions. All images are normalized using ImageNet mean and standard deviation statistics to match the pretrained encoder’s input distribution and then converted to PyTorch tensors.

d) FULL-RESOLUTION EVALUATION

In addition to cropped validation, we also report full-resolution evaluation by running inference on the complete tile with deterministic preprocessing (normalization only, no cropping).

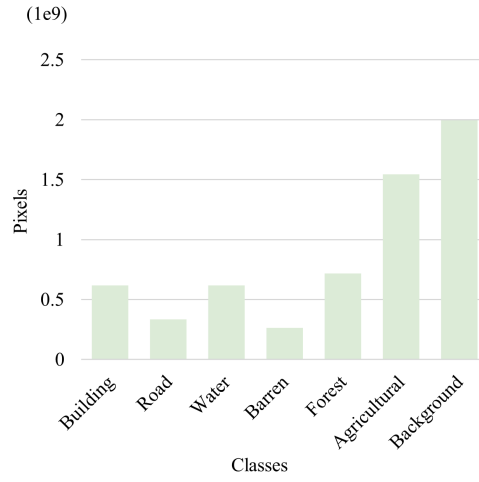


Figure 2: Pixel-level class distribution in the LoveDA dataset

3.2. DEEP NEURAL NETWORK ARCHITECTURE: DEEPLABV3+

The core of the classification framework is the DeepLabV3+ architecture. This model represents the culmination of the DeepLab series, introducing an Encoder-Decoder structure that combines the semantic richness of ASPP with the spatial precision of a decoding pathway.

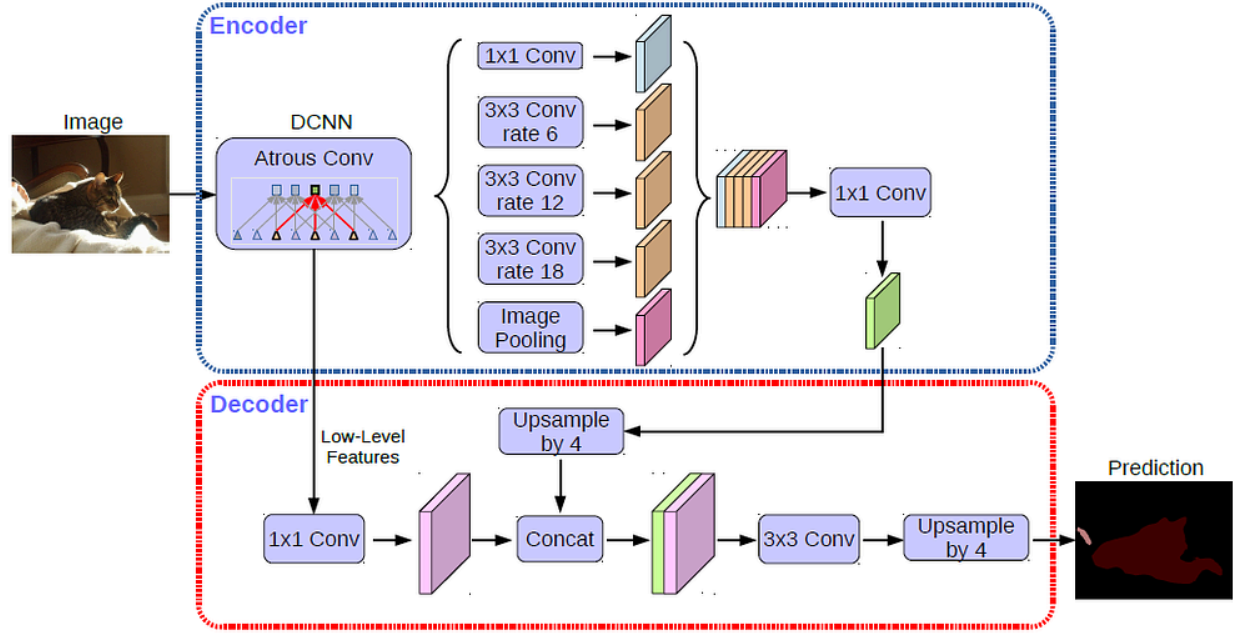


Figure 3: DeepLabV3+ model architecture

At its foundation, DeepLabV3+ utilizes a backbone network (e.g., ResNet-101, ResNet152) to extract deep hierarchical features from input images. These features are then processed by the Atrous Spatial Pyramid Pooling(ASPP) module, which captures multi-scale context by applying atrous(dilated) convolutions with different dilation rates in parallel. This enables the network to aggregate features at multiple receptive fields, making it highly effective for recognizing objects and patterns of varying sizes without increasing the number of parameters or losing resolution.

What sets DeepLabV3+ apart is its encoder-decoder structure. While the encoder captures semantic features using ASPP, the decoder progressively upsamples the encoded feature maps to recover sharp object boundaries. This dual-pathway design addresses the common trade-off in CNNs between high-level semantic abstraction and low-level spatial detail. The decoder uses skip connections to fuse low-level features from earlier layers of the backbone, helping preserve edge information and improve segmentation accuracy around complex boundaries.

3.3. ENCODER BACKBONE: RESNET-152

In this study we utilize the ResNet-152 (Residual Network with 152 layers) as the feature extraction backbone. The choice of ResNet-152 over shallower networks such as ResNet-50 or ResNet-101 is motivated by the need for deeper semantic abstraction, which is crucial for differentiating between spectrally and spatially similar land-cover classes in high-resolution remote sensing imagery. We conducted preliminary experiments with different backbones, including ResNet-50, ResNet-101, Xception and EfficientNet-B7. ResNet-152 was selected as the final backbone as it demonstrated better performance in capturing complex spatial features.

A) RESIDUAL BUILDING BLOCKS

The network is composed of stacked residual blocks, each designed to mitigate the vanishing gradient problem in very deep networks. A residual block performs the computation:

$$y = F(x, \{W_i\}) + x$$

Where x is the input, F is the residual mapping and y is the output. These skip connections allow gradients to propagate directly through identity mappings, stabilizing training in deep architectures like ResNet-152.

B) BOTTLENECK DESIGN

Each residual block adopts a bottleneck architecture comprising a sequence of:

- 1×1 convolution (for dimensionality reduction),
- 3×3 convolution (for spatial processing),
- 1×1 convolution (for restoring dimensions).

This design significantly reduces the number of parameters and computation while enabling the depth required for rich semantic feature learning.

By employing ResNet-152, the encoder gains deeper hierarchical representations, which are essential for robust pixel-level classification in complex and heterogeneous land-cover scenes.

3.4 EBCT: ENTROPY-BASED CURRICULUM TRAINING STRATEGY

To move beyond uniform sampling, we introduce **Entropy-Based Curriculum Training (EBCT)**, a simple training wrapper that orders samples from *easy* to *hard* using the model's predictive uncertainty. The core idea is to (i) score each training image with an **entropy-based difficulty measure**, (ii) sort images from low to high difficulty, and (iii) train on an increasing fraction of the sorted list over epochs. EBCT does **not** modify the network architecture; it only changes which training images are included at each epoch.

A) PIXEL-WISE PREDICTIVE ENTROPY

Given the model's softmax probabilities $p_c(i)$ over C classes at pixel i , we define the Shannon entropy:

$$H(i) = - \sum_{c=1}^C p_c(i) \cdot \log(p_c(i))$$

where probabilities are numerically stabilized (e.g., clamped away from zero) to avoid $\log(p_c(i))$.

B) IMAGE-LEVEL DIFFICULTY SCORE

For each training image, we estimate an image-level difficulty score by computing the mean pixel entropy on a small set of **deterministically sampled crops**. Concretely, we sample K crops of fixed size from the image (using a seed derived from the global seed, epoch index, crop index, and a stable image identifier), run the current model in evaluation mode to obtain per-pixel class probabilities, compute the mean entropy for each crop, and finally average across crops:

$$D(x) = \frac{1}{K} \sum_{k=1}^K \left(\frac{1}{P_k} \right) \sum_{i \in P_k} H_k(i)$$

where P_k denotes the pixel set of crop k . Entropy scoring is performed **without gradients** and uses **deterministic preprocessing** (normalization only), avoiding stochastic augmentations so that difficulty estimates remain stable.

C) CURRICULUM SCHEDULE

Let N be the number of training images and let the images be sorted by difficulty $D(x)$ in ascending order (lowest entropy = easiest). At epoch t , EBCT selects the first $\lceil p_t \cdot N \rceil$ images from this sorted list, where p_t is the curriculum fraction. We use a short **warm-up** stage with the full training set to stabilize early optimization, and then linearly expand the subset size from an initial fraction to the full dataset:

- Warm-up: for $t < T_w$, use all training images.
- Curriculum: for $T_w \leq t \leq T_{end}$, increase p_t linearly from p_{start} to $p_{end} = 1.0$.
- After curriculum: for $t > T_{end}$, use all training images.

To account for the fact that model confidence evolves during training, we **recompute** image difficulty scores and re-sort the dataset at a fixed interval (every R epochs). Between recomputation steps, the last computed ordering is reused.

D) DATASET-SPECIFIC SETTINGS

We use $p_{start} = 0.5$, $p_{end} = 1.0$, $T_w = 5$ warm-up epochs, and a recomputation interval of $R=5$ epochs for both datasets. For entropy estimation, we use $K=2$ deterministic crops per image on **DeepGlobe** because each tile is 2448×2448 and scoring the full image would be unnecessarily expensive; sampling two 1024×1024 crops provides a cheap but sufficiently representative difficulty estimate. In contrast, **LoveDA** tiles are fixed at 1024×1024 and our training already operates on a single 768×768 crop, so $K = 1$ deterministic crop per image is sufficient for stable entropy-based ranking.

The linear ramp reaches the full dataset by epoch **35** on DeepGlobe and by epoch **30** on LoveDA. We set these end epochs to (i) allow the difficulty ordering to stabilize after warm-up, while (ii) reserving enough remaining epochs for optimization on the full dataset. DeepGlobe uses a slightly longer ramp because difficulty estimates from large tiles are inherently noisier when approximated by a small number of crops, making early inclusion of high-entropy samples more sensitive; LoveDA reaches the full dataset earlier to avoid under-coverage given its larger sample count and more stable per-tile difficulty estimates.

E) CONTROLLED COMPARISON

All other components (patch extraction, boundary-guided sampling where applicable, augmentations, losses, and optimization) are kept identical between baseline and EBCT runs; EBCT only changes the **epoch-wise subset of training images** used by the dataloader.

3.5 TRAINING

All experiments were implemented in PyTorch using the segmentation-models-pytorch framework. Baseline and EBCT runs share the same architecture, losses, optimizer, scheduler, augmentations, and total training budget; the only difference is the data selection policy per epoch (uniform sampling for baseline vs. curriculum-based subset selection for EBCT). Training is performed on mini-batches of fixed-size image patches to ensure stable GPU memory usage.

3.5.1 COMMON INPUT PIPELINE

Training is performed on tiled patches extracted from high-resolution imagery. For **DeepGlobe**, each 2448×2448 tile is sampled into 1024×1024 patches (two patches per image per epoch). For **LoveDA**, images are already provided as 1024×1024 tiles and a single 768×768 crop is used per sample during training, which is sufficient to capture local context while improving augmentation diversity and memory efficiency. Inputs are normalized using ImageNet mean and standard deviation to match the pretrained ResNet-152 encoder statistics and then converted to PyTorch tensors.

Ground-truth masks are represented as **single-channel class-index maps**. Dataset-specific ignore regions (e.g., no-data/unknown) are excluded from loss and metric computation via an ignore label.

3.5.2 TRAINING SETUP

Training is performed with:

- Optimizer: AdamW
- Batch Size: 8
- Loss: Dice Loss + Multiclass Cross-Entropy
- Scheduler: Polynomial (Poly) LR schedule
- Poly Power: 0.9
- Epochs: 50
- Weight Decay: $2e-4$
- Initial Learning Rate: $2e-4$
- Minimum Learning Rate: $5e-6$
- Checkpointing: Best model selected by the highest val. mIoU

All parameters are updated using backpropagation and validation loss is computed at the end of each epoch. This baseline phase establishes a stable model whose predictions are then used for entropy estimation in the next stage.

3.5.3 BASELINE VS. EBCT TRAINING REGIMES

Baseline training uses uniform sampling over the entire training set for all epochs.

EBCT training follows the same optimization procedure but changes the data exposure schedule: after a short warm-up using the full training set, the model is trained on an increasing fraction of samples ordered by entropy-based difficulty. Difficulty scores are recomputed periodically, and the subset size is expanded linearly until reaching the full dataset; training then continues on the full dataset for the remaining epochs.

3.5.4 MONITORING, VALIDATION, AND CHECKPOINTING

We log training and validation curves using TensorBoard, including total loss and segmentation quality metrics (mIoU/IoU, F1-score, and pixel accuracy). Model checkpoints are saved at the end of each epoch, and the **best checkpoint** is selected by the **highest validation mIoU**. Early stopping is disabled to ensure that EBCT completes its planned curriculum schedule under a fixed training budget.

3.5.5 HYPERPARAMETER SELECTION

Hyperparameters were selected through empirical tuning on the validation split. The batch size was capped at 8 due to GPU memory constraints when training on large patches. A training budget of 50 epochs was chosen as a practical trade-off: shorter schedules tended to underfit, while longer runs yielded diminishing returns. EBCT parameters were set to include a **5-epoch warm-up**, followed by a **linear curriculum ramp** that reaches the full dataset by **epoch 35 on DeepGlobe** and **epoch 30 on LoveDA**, leaving the remaining epochs for standard full-data optimization.

3.5.6 REGULARIZATION AND ROBUSTNESS MEASURES

To reduce overfitting and improve generalization, we apply strong data augmentation during training (spatial flips/rotations and mild photometric jitter), while validation uses deterministic preprocessing (fixed cropping and normalization without stochastic augmentations). We additionally report both cropped and full-resolution evaluation protocols where applicable to assess robustness under realistic inference settings.

3.6 EVALUATION METRICS

3.6.1 DICE LOSS

Based on the Dice Similarity Coefficient, this loss directly optimizes the intersection-over-union, making it robust to class imbalance.

$$L_{dice} = \frac{2 \cdot \sum p_{true} \cdot p_{pred}}{\sum p_{true}^2 + \sum p_{pred}^2 + \epsilon}$$

where:

- ϵ is a smoothing factor
- P_{true} represents the ground truth binary mask
- P_{pred} represents the predicted probability map

3.6.2 IOU SCORE (INTERSECTION-OVER-UNION)

Intersection over Union (IoU), also known as the **Jaccard Index**, is a widely used metric to evaluate the overlap between the predicted segmentation and the ground truth in semantic segmentation tasks. It quantifies how well the predicted class regions align with the actual ones.

The IoU is calculated as:

$$IoU = \frac{TP}{TP + FP + FN}$$

IoU values range from 0 to 1, where a higher IoU indicates better alignment between the prediction and ground truth. Scores above 0.5 are typically considered acceptable, while values above 0.7 suggest strong segmentation performance. Unlike simple accuracy, IoU takes both false positives and false negatives into account, making it a more robust metric, especially in datasets with class imbalance.

3.6.3 ACCURACY

Accuracy is the overall correctness of the model and measures the ratio of correctly predicted instances to the total number of predictions. While useful, accuracy can be misleading in imbalanced datasets, hence it is often reported alongside metrics like IoU, F1 Score and Dice Loss.

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN}$$

3.6.4 RECALL

Recall, also known as Sensitivity or True Positive Rate, measures the ability of the model to identify all relevant instances(i.e., how many actual positives were correctly predicted). High recall means most of the positive samples are correctly detected.

$$Recall = \frac{TP}{TP + FN}$$

High recall means most of the positive samples are correctly detected.

3.6.5 PRECISION

Precision measures the proportion of true positive predictions among all predicted positives.

$$Precision = \frac{TP}{TP + FP}$$

High precision indicates a low false positive rate.

3.6.6 F1 SCORE

F1 Score is the harmonic mean of Precision and Recall. It provides a balanced measure that takes both false positives and false negatives into account, especially useful in cases of class imbalance.

$$F1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall}$$

A high F1 score indicates that the model achieves both high recall and high precision.

3.7 COMPUTE / EFFICIENCY MEASUREMENT

To quantify the computational efficiency of EBCT, we report training-time cost using lightweight, implementation-aligned proxies rather than hardware-dependent FLOPs. We track compute in two complementary units: **image computations** (patch-level network evaluations) and **pixel throughput** (processed pixels).

3.7.1 IMAGE COMPUTATIONS

We define **image computations** as the total number of patch-level passes through the network during training. This includes (i) standard training passes (forward + backward) and (ii) additional forward-only passes used for EBCT entropy recomputation.

For an epoch t , let U_t be the number of optimizer updates (i.e., the number of mini-batches) and B be the batch size. Then the number of training patches processed that epoch is:

$$N_t^{train} = U_t \cdot B$$

We count one forward and one backward per training patch, so:

$$TrainFwd_t = N_t^{train}, TrainBwd_t = N_t^{train}.$$

EBCT additionally recomputes difficulty scores at a fixed interval (every R epochs after warm-up). On recomputation epochs, we perform forward-only inference to estimate entropy-based difficulty. If R is the number of training images and K is the number of deterministic crops used per image for entropy estimation, then:

$$\begin{aligned} EntropyFwd_t \\ = N \cdot K, \text{ if entropy is recomputed at epoch } t \\ = 0, \text{ otherwise} \end{aligned}$$

Finally, total image computations over the full run are computed as:

$$TotalCompute = \sum_t (TrainFwd_t + TrainBwd_t + EntropyFwd_t)$$

To make EBCT’s additional cost explicit, we report an overhead ratio:

$$EntropyOverTrain = \frac{\sum_t EntropyFwd_t}{\sum_t (TrainFwd_t + TrainBwd_t)}$$

This isolates the compute introduced purely by entropy recomputation, independent of the compute saved by training on smaller subsets early in the curriculum.

3.7.2 PIXEL THROUGHPUT (SECONDARY PROXY)

Since convolutional compute scales strongly with the number of processed pixels, we also report the total number of pixels processed during training and entropy estimation. Let S be the crop size (in pixels). For epoch t :

$$SeenPixels_t = N_t^{train} \cdot S^2, EntropyPixels_t = EntropyFwd_t \cdot S^2,$$

and:

$$TotalPixels = \sum_t (SeenPixels_t + EntropyPixels_t).$$

3.7.3 DATASET-SPECIFIC SETTINGS USED IN COMPUTE ACCOUNTING

- **DeepGlobe:** Training operates on 1024×1024 patches and uses two crops per image per epoch due to the 2448×2448 tile size. Entropy estimation uses $K = 2$ deterministic crops per image.
- **LoveDA:** Training operates on 768×768 crops sampled from 1024×1024 tiles; a single crop is sufficient per image for entropy estimation, so $K = 1$.

3.7.4 REPORTING

We report (i) total image computations, (ii) entropy overhead ratio, and (iii) pixel throughput for both baseline and EBCT. All measurements are computed consistently across runs; baseline uses uniform sampling over the full training set, whereas EBCT changes only the epoch-wise subset size according to the curriculum schedule.

3.8 HARDWARE

Experiments were conducted on Google Colab using Nvidia A100 Tensor Core GPUs. The A100 GPU provides high-throughput tensor computation and large on-board memory, which is critical for training deep semantic segmentation networks. This hardware configuration enabled efficient training of the DeepLabV3+ architecture with large input patches and supported the additional computational overhead introduced by the entropy-based curriculum sampling procedure. All models were trained using mixed-precision acceleration (PyTorch AMP) to further reduce memory consumption and improve training speed without loss of numerical stability.

3.9 ETHICS

No human participants or personal data were used. The study relies solely on publicly available satellite imagery datasets; therefore, no consent or confidentiality procedures were required.

4. EXPERIMENTS AND RESULTS

4.1 EXPERIMENTAL SETUP

All experiments were conducted using the PyTorch deep learning framework on a Linux-based system running Ubuntu 22.04.4 LTS (Jammy Jellyfish). The software environment consisted of Python 3.12.12 and PyTorch 2.9.0 with CUDA 12.6 support. Training was performed on a high-performance computing setup equipped with a single NVIDIA A100 GPU with 40 GB of VRAM and 83.5 GB of system memory, which enabled efficient processing of high-resolution remote sensing imagery. The implementation was executed in a Google Colab environment using its default system configuration, which is inherently Linux-based.

4.2 EXPERIMENT 1: EBCT VS BASELINE ON DEEPGLOBE

A) PURPOSE

The primary objective of this experiment was to evaluate the effectiveness of the proposed Entropy-Based Curriculum Training (EBCT) strategy compared to a standard training baseline on the DeepGlobe Land Cover dataset. Our hypothesis is that ordering training samples from easier (low-entropy) to harder (high-entropy) examples stabilizes optimization and yields better performance.

B) SETUP AND DATA

We trained DeepLabV3+ with a ResNet-152 encoder under two conditions: (i) Baseline training using uniform sampling and (ii) EBCT training using entropy-based curriculum scheduling. Both models were trained with identical settings (identical optimizer, learning rate,

weight decay, scheduler, batch size and total number of epochs) and the same preprocessing pipeline. Data was split into 90% training and 10% validation sets.

C) KEY FINDINGS

Figure 4 illustrates the mean Intersection-over-Union (mIoU) scores obtained across five independent runs with different random seeds. EBCT consistently outperforms the baseline in all runs and exhibits lower variance, indicating a more stable training behavior. While the absolute improvements are modest, the reduced fluctuation across runs suggests that entropy-guided curriculum learning leads to more reliable convergence.

Figure 5 presents the class-wise IoU comparison averaged over all runs. EBCT improves performance in most categories, demonstrating its effectiveness in learning dominant land-cover patterns. However, the method does not yield uniform gains across all classes. Forest, which is visually more complex and spectrally similar, shows slight degradation. This suggests that while EBCT facilitates learning from easier and moderately complex samples, high-entropy classes remain challenging and may require longer training schedules or more frequent entropy updates.

Overall these results indicate that EBCT provides a more stable optimization process and selective class-wise improvements, while highlighting the need for additional mechanisms to better exploit difficult samples.

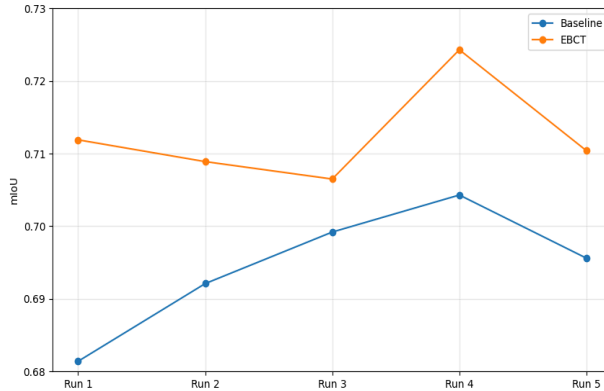


Figure 4: Baseline vs EBCT results across different random seeds

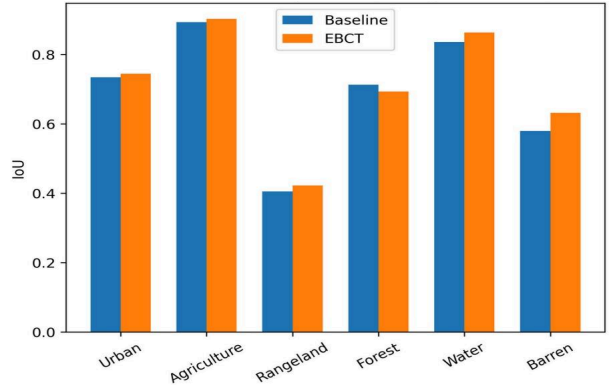


Figure 5: IoU comparison of each class

4.3 EXPERIMENT 2: EBCT VS BASELINE ON LOVE DA

A) PURPOSE

This experiment tests whether the benefits of EBCT generalize to LoveDA, a dataset explicitly designed to include domain shift between Urban and Rural scenes. The hypothesis is that EBCT improves robustness by first learning stable patterns from low-entropy samples and gradually including high-entropy, domain-specific boundary regions.

B) SETUP AND DATA

We repeated the same two training configurations: Baseline and EBCT, using LoveDA’s official train/validation split and the identical training setup described previously. Evaluation was performed using the same indicators. Where applicable, results are also reported separately for the Urban and Rural domains.

C) KEY FINDINGS

Figure 6 presents the mIoU scores across five independent runs. EBCT consistently achieves slightly higher mIoU values than the baseline in all runs, while also exhibiting reduced fluctuations. Although the absolute performance gains are modest, the consistent improvement across different random seeds indicates that EBCT leads to a more stable and reliable training process.

Figure 7 presents the class-wise IoU averaged over all runs. EBCT improves performance in several categories, which suggests that the curriculum strategy facilitates learning of dominant and moderately complex land-cover patterns. However, EBCT does not outperform the baseline uniformly across all classes. These results indicate that while EBCT effectively enhances overall performance and robustness, highly structured or spectrally ambiguous classes remain challenging.

Overall, the LoveDA experiment demonstrates that EBCT’s gains are not limited to a single benchmark and remain effective under domain shift. Together with Experiment 1, these findings suggest EBCT is a practical training-time enhancement that can improve segmentation performance with a favorable accuracy-efficiency trade-off.

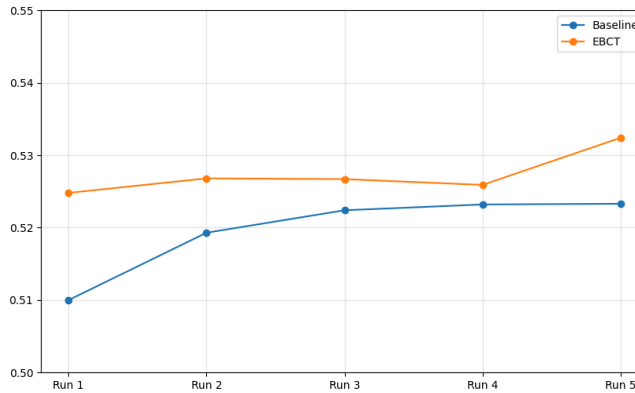


Figure 6: Baseline vs EBCT mIoU across different random seeds

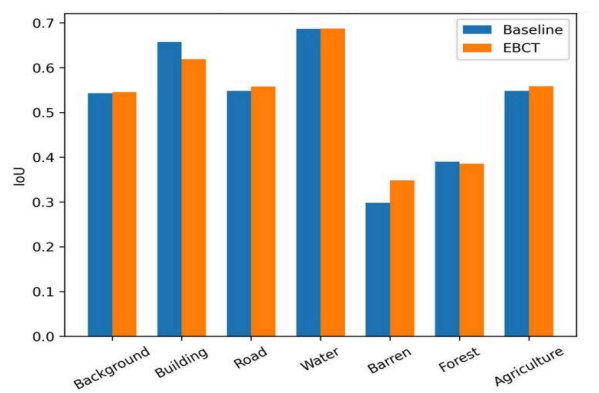


Figure 7: Class-wise IoU comparison

A key observation regarding EBCT’s behavior, particularly on the LoveDA dataset, is its accelerated convergence. While the baseline model required more epochs to stabilize its predictions on urban-rural boundaries, EBCT reached its optimal validation mIoU significantly earlier (e.g., at epoch 7). This suggests that the curriculum-driven approach acts as an implicit

optimizer, guiding the network toward a stable local minimum more directly by reducing the gradient noise typically introduced by 'hard' samples early in training. This efficiency makes EBCT a practical solution for time-sensitive or large-scale mapping projects.

4.4 EFFICIENCY RESULTS

Experiments on the DeepGlobe dataset were first conducted using a fixed random seed (seed = 42) to establish a stable baseline configuration. Subsequently, additional runs were performed using randomly sampled seeds drawn uniformly from the range [0, 9999] to assess robustness against random initialization effects. For each seed, both the baseline training strategy and the proposed EBCT approach were evaluated under identical training and inference conditions.

Across multiple seeds, EBCT consistently achieved comparable or slightly higher mean Intersection-over-Union (mIoU) scores than the baseline. For instance, with seed = 7366, EBCT improved the final validation mIoU from 0.699 to 0.706, while with seed = 217, the improvement was more pronounced, increasing mIoU from 0.692 to 0.709. Similar trends were observed for seed = 1558, where EBCT achieved an mIoU of 0.710 compared to the baseline’s 0.696. These results indicate that EBCT provides a modest but consistent performance gain that is stable across different random initializations.

A detailed class-wise analysis revealed that EBCT particularly benefits visually ambiguous and minority classes such as rangeland, barren land, and water. Confusion matrix evaluations show reduced misclassification between agriculture–rangeland and rangeland–forest boundaries under EBCT, especially in the final evaluation setting. While agriculture and forest classes already exhibit high recognition accuracy under the baseline, EBCT improves boundary delineation by reallocating training emphasis toward higher-entropy regions during later epochs, leading to improved class balance without degrading performance on dominant classes.

Random seed	Baseline mIoU	EBCT mIoU	Δ mIoU (EBCT–Base) $\uparrow$
42	0.6814	0.7119	0.0305
217	0.6921	0.7089	0.0168
7366	0.6992	0.7065	0.0073
1881	0.7043	0.7243	0.0200
1558	0.6956	0.7104	0.0148
Mean $\pm$ Std	0.6945 $\pm$ 0.0086	0.7124 $\pm$ 0.0069	0.0179 $\pm$ 0.0085

Table 2: DeepGlobe Land Cover: Baseline vs. EBCT comparison across random seeds.

Random seed	Baseline mIoU	EBCT mIoU	Δ mIoU (EBCT–Base) $\uparrow$
42	0.5100	0.5248	0.0148
445	0.5193	0.5268	0.0075
96	0.5224	0.5267	0.0043
2478	0.5232	0.5259	0.0027
227	0.5233	0.5324	0.0091
Mean $\pm$ Std	0.5196 $\pm$ 0.0056	0.5273 $\pm$ 0.0030	0.0077 $\pm$ 0.0047

Table 3: LoveDA: Baseline vs. EBCT comparison across random seeds.

To further evaluate generalization and computational efficiency, experiments were conducted on the LoveDA dataset using a randomly selected seed (seed = 2478). The dataset was split into 2,522 training images, 1,669 validation images, and 1,796 test images. EBCT reached its best validation performance early in training, with the optimal checkpoint achieved at epoch 7, yielding a validation mIoU of 0.5259.

Compared to baseline training, EBCT significantly reduced the total number of forward and backward training image computations. Specifically, EBCT processed approximately 219k training images, compared to 252k for the baseline, corresponding to a 8% reduction in training computations. Although EBCT introduces additional entropy estimation overhead, the total computational workload remained lower, with EBCT requiring only 91.9% of the baseline’s total image computations. These results demonstrate that EBCT not only improves or maintains segmentation performance but also enhances training efficiency, making it particularly suitable for large-scale, high-resolution datasets such as LoveDA.

Method	Training compute	Entropy overhead	Total	Ratio
Baseline	252000	0	252000	1.000
EBCT	219104	12610	231714	0.919

Table 4: Analysis of Computational Efficiency and Training Overhead.

The marginal gains in mIoU should be evaluated alongside the computational savings presented in Table 3. In deep learning for Earth Observation, where datasets are massive (e.g., DeepGlobe and LoveDA), a reduction of $\sim 8\%$ in forward and backward passes without sacrificing accuracy is significant. EBCT demonstrates that by prioritizing 'informative' samples through entropy-based ranking, the model reaches a superior optimization state faster than uniform sampling, effectively 'doing more with less'.

5. CONCLUSION

This study investigated the effectiveness of Entropy-Based Curriculum Training (EBCT) as a training-time enhancement for high-resolution land-cover semantic segmentation. Using DeepLabV3+ as a strong baseline, experiments were conducted on two public benchmarks, DeepGlobe and LoveDA, to assess both performance gains and computational behavior.

The results demonstrate that EBCT consistently yields modest but stable improvements over uniform sampling, achieving approximately 1-1.5% gains in mIoU (mean intersection over union) across both datasets. Crucially, these findings establish EBCT as a resource-aware training paradigm rather than a purely performance-driven modification. By intelligently prioritizing samples based on their information gain, EBCT achieves a superior optimization state while simultaneously reducing overall computational costs by approximately 8%. This 'learning smarter' approach suggests that the semantic gap in high-resolution imagery can be bridged more effectively by following a structured optimization trajectory that minimizes redundant gradients from well-learned, low-entropy regions. These findings support the hypothesis that entropy-guided sample ordering stabilizes early optimization and encourages the network to first learn robust representations from visually homogeneous regions before being exposed to more complex, ambiguous samples.

Further analysis indicates that EBCT provides most noticeable benefits on low-entropy ("easy") samples, while performance gains on high-entropy ("hard") regions remain more limited, sometimes even worse. This suggests that while curriculum-driven training improves convergence behavior and efficiency, it may require longer refinement phases.

Several limitations should be acknowledged. The curriculum schedule and entropy recomputation frequency were fixed and no extensive hyperparameter search was performed. Moreover improvements, although consistent, remain moderate, highlighting the inherent difficulty of high-resolution land-cover segmentation.

Future work could investigate more frequent entropy recomputation strategies to better capture the evolving difficulty of training samples. Additionally extended training schedules with a higher number of epochs may further refine representations on high-entropy regions; however this must be carefully balanced against EBCT's advantage in reducing the computational cost.

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APPENDIX

APPENDIX A. CODE AVAILABILITY

The code used for preprocessing, training, and evaluation is publicly available at [this github link](#).

APPENDIX B. STEPS PER EPOCH GRAPH

Figure B1 illustrates the number of optimizer update steps performed per epoch under baseline training and EBCT. In the baseline configuration, the full training set is used throughout all epochs, resulting in a constant number of optimizer updates per epoch. In contrast, EBCT dynamically adjusts the effective training set size during the curriculum phase by gradually increasing the number of samples from low-entropy to high-entropy samples.

As shown in Figure B1, EBCT performs substantially fewer optimizer updates during the early curriculum epochs, reflecting training on a reduced subset of easier samples. The number of steps then increases approximately linearly as additional high-entropy samples are introduced, until it matches the baseline level once the full dataset is incorporated.

This behavior directly demonstrates the computational efficiency of EBCT: fewer optimizer updates are required during the early and intermediate stages of training, reducing the total number of processed image patches and associated GPU computations. At the same time, the final training phase ensures that the model is still exposed to the complete data distribution. This appendix therefore provides quantitative evidence supporting EBCT’s favorable accuracy-efficiency trade-off.

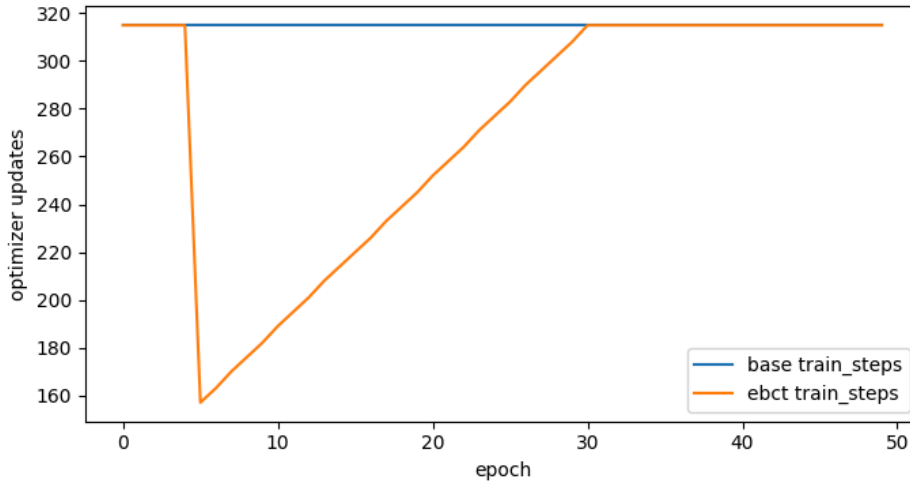


Figure B1. Comparison of Training Steps per Epoch between Baseline and EBCT

APPENDIX C. CONFUSION MATRIX ANALYSIS

Figure C1 presents row-normalized confusion matrices for the baseline model and EBCT under the same experimental configuration. Each row corresponds to a ground-truth class, and each cell represents the proportion of pixels from that class predicted as a given category. Diagonal values therefore indicate per-class recall, while off-diagonal values highlight dominant confusion patterns.

Compared to the baseline, EBCT exhibits stronger diagonal dominance in several classes, indicating improved class-specific discrimination.

Despite these gains, both models continue to exhibit confusion between structurally and spectrally similar categories, reflecting the intrinsic difficulty of these classes in high-resolution satellite imagery. This supports the quantitative findings that EBCT improves overall robustness and class-wise performance, while hard boundary regions and heterogeneous land-use patterns remain challenging.

Overall this confusion matrix analysis complements the mIoU and class-wise IoU results by demonstrating that EBCT's improvements arise not only from marginal score increases, but also from a reduction in systematic misclassification between key land-use categories.

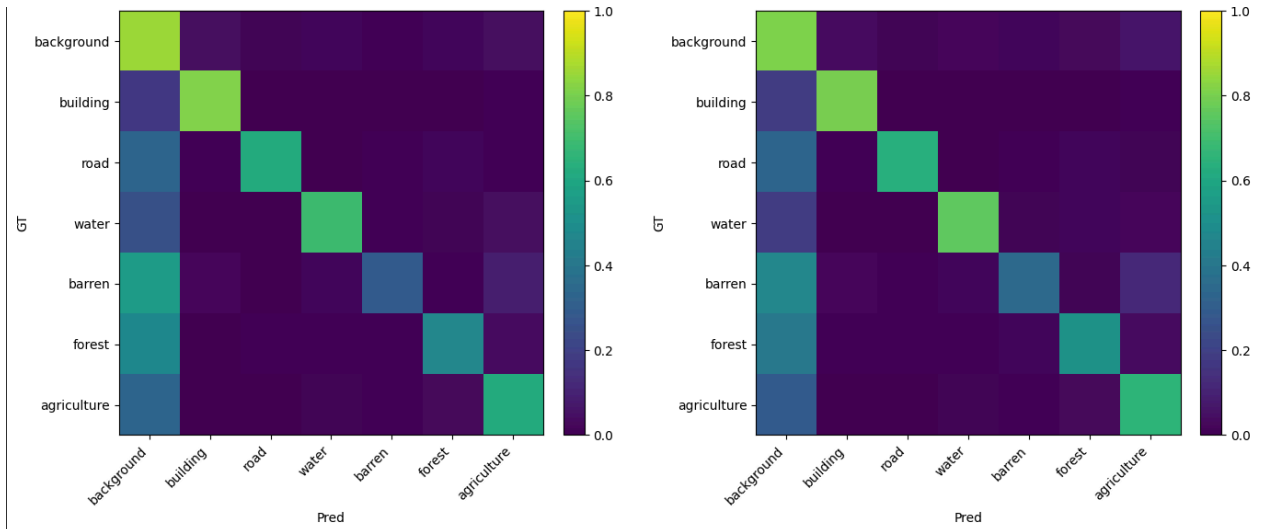


Figure C1. Row-Normalized Confusion Matrices (Baseline on the left, EBCT on the right)

APPENDIX D. QUALITATIVE SEGMENTATION RESULTS

Figure D1 presents representative qualitative examples comparing the input satellite images, ground-truth masks, and the corresponding model predictions. These visualizations provide insight into the spatial behavior of the segmentation model beyond aggregate quantitative metrics.

As illustrated in the examples, the model successfully captures the large-scale structure of dominant land-cover classes demonstrating strong semantic consistency at the region level. Linear structures such as major roads are generally well preserved, indicating that the encoder-decoder architecture effectively models long-range spatial context.

At the same time, the predictions reveal typical failure modes of high-resolution land-cover segmentation. Errors are predominantly concentrated along class boundaries and small fragmented objects, where mixed spectral responses and fine-grained textures increase ambiguity.

These qualitative observations are consistent with the quantitative and confusion-matrix analyses, confirming that most errors arise from heterogeneous transition zones and spectrally overlapping classes rather than from large homogeneous regions. This appendix therefore complements the numerical evaluation by highlighting the spatial characteristics of model successes and limitations.

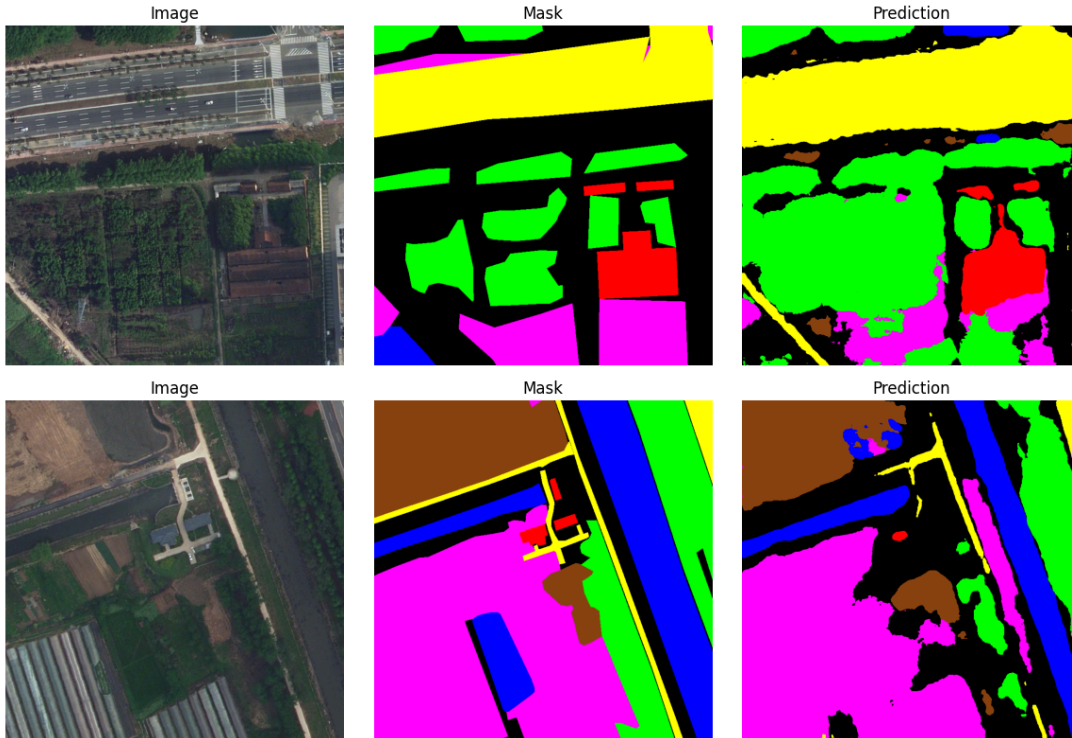


Figure D1. Qualitative segmentation examples. Input satellite image, corresponding ground-truth annotation and the predicted segmentation map.